Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Multi-omics biomarker analysis for longitudinal treatment monitoring and prediction			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1 (**Large Category**)**		**Technology Field 2 (**Large Category**)**	
	Biomedical and Healthcare			
	Medium Category	Small Category	Medium Category	Small Category
	Therapeutic and diagnostic systems	*Molecular diagnostic systems*		
Concept of Proposed Technology	AI-powered, integrated multi-omics molecular analysis platform technology to monitor responses to anti-cancer therapy and predict clinical outcomes. • Spatial transcriptomics, DNA/RNA/proteomics analysis, or highly-throughput single-cell analysis technologies on solid tumor tissues • Liquid biopsy technologies for circulating biomarkers, such as circulating tumor cells, cell-free DNA/RNA, extracellular vesicles, among others • Cross-modality data integration and AI/ML modeling to derive predictive signatures of response, resistance, and minimal residual disease			
Rationale for Joint R&D Project	• *Clinical unmet need: In cancer therapy, close monitoring for emerging resistance and robust prediction of short- and long-term outcomes are critical to enable timely treatment decision and improve survival.*			

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	• *Technical unmet need: Along the advances in next-generation targeted therapies, accurate molecular analysis on dynamic changes of tumors and their microenvironments are imperative to capture dynamic molecular changes in real-time during therapy.* • *A joint R&D effort between a world-leading biomedical/clinical research institution and an experienced commercial partner is essential to de-risk product development, execute rigorous clinical validation, and accelerate commercialization and clinical translation.*
Key Words	1. Longitudinal treatment monitoring and prediction 2. Multiomics analysis 3. Liquid biopsies 4. Biomarkers

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (5)
R&D Objectives	• *Develop an AI-powered multi-omics platform that enables high-throughput molecular analysis for monitoring longitudinal treatment response monitoring and predicting clinical outcomes* • *Identify a set of biomarkers for a standard anti-cancer therapy currently used in clinics* • *Provide a clinically applicable prototype for clinical validation and applications*			
R&D Target Specifications	• *Technologies or biomarkers applicable to at least two or more clinically approved anti-cancer drugs* • *Multi-omics approach integrating at least three different types of biomarkers on both tissue and easily assessable biofluids* • *Early nonresponse prediction: AUROC ≥0.85 using longitudinal features (baseline and during treatment).* • *Clinical classification (response vs. nonresponse): sensitivity ≥80% and specificity ≥80% in an independent test set.*			
R&D Plan	• *Assess the company's current molecular analysis technology and clinical portfolio to evaluate the clinical feasibility in the US and global setting.* • *Establish the next developmental target disease and drugs* • *Define molecular targets in multiomics analysis* • *Develop and validate analytical technologies compared to currently available competitive solutions* • *Conduct independent clinical validation study to evaluate the technology's accuracy*			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *Assess the company's current molecular analysis technology for clinical applications in the US and global markets.* • *Develop drug-resistant model systems for technology development and validation* • *Develop and validate one of the analytical sensing technology for multi-omics analysis* • *Assist in ML/AI development through collaboration with Korean SME partner*
Expected R&R of Korean SME Partner	• *Define the target disease and drug therapy for the company's next commercial pipeline* • *Conduct market analysis for the proposed multiomics analysis technology for therapeutic monitoring and prediction* • *Develop AI/ML-based multiomics analysis platforms* • *Design a clinical study with either retrospective or prospective cohorts.* • *Validate multiomics analysis platform using model systems and serially collected clinical samples*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	• *Validated multiomics analysis workflow for clinical specimens* • *Clinically relevant biomarker signatures for standard anti-cancer therapies* • *AI/ML-based prototype system for therapeutic monitoring and prediction* • *Clinical evidence for cross-therapy applicability with clinical validation results.* • *Clinical impact: Earlier identification of nonresponders and emerging resistance enables timely treatment modification, reduces exposure to ineffective therapy/toxicity, and clinical outcomes* • *Technical impact: Integrated tissue–liquid biopsy multi-omics will yield mechanistic insights into resistance biology and tumor–microenvironment dynamics, generating actionable biomarker signatures* • *Commercial impact: The new pathway of monitoring resistance and clinical outcomes for standard anti-cancer drugs in major clinics will open up new avenues of global market and early penetration to major hospitals*